

Supplementary Appendix

Causal Emergence in Financial Markets: Dynamic Organization and Effective Dimensionality During Systemic Stress

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1 Scale-Invariance Numerical Test and Formal Derivation

A central property of the empirical estimator developed in this paper is exact invariance to common/global scalar rescaling of asset return units ($X \rightarrow cX$ for $c > 0$). This ensures that whether returns are expressed in decimals, percentages, or basis points, the information-theoretic quantities CEFI_t and q_t^* remain numerically identical.

1.1 Mathematical Derivation of Scale Invariance

Consider the transition dynamics $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \boldsymbol{\varepsilon}_{t+1}$ with $\boldsymbol{\varepsilon}_{t+1} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon)$ and unconditional state covariance $\boldsymbol{\Sigma}_x = \text{Cov}(\mathbf{x}_t)$. Under a global scalar change of return units $\tilde{\mathbf{x}}_t = c\mathbf{x}_t$ ($c > 0$):

1. **State Covariance:** $\tilde{\boldsymbol{\Sigma}}_x = \text{Cov}(c\mathbf{x}_t) = c^2\boldsymbol{\Sigma}_x$.
2. **Transition Matrix Estimator:** Under the trace-scaled ridge estimator,

$$\tilde{\lambda}_t = \lambda_0 \cdot \frac{\text{Tr}(\tilde{\mathbf{X}}_{\text{lag}}^\top \tilde{\mathbf{X}}_{\text{lag}})}{p} = c^2\lambda_t \quad (1)$$

Consequently,

$$\hat{\tilde{\mathbf{A}}} = \tilde{\mathbf{X}}_{\text{lead}}^\top \tilde{\mathbf{X}}_{\text{lag}} \left(\tilde{\mathbf{X}}_{\text{lag}}^\top \tilde{\mathbf{X}}_{\text{lag}} + \tilde{\lambda}_t \mathbf{I}_p \right)^{-1} = c^2 \mathbf{X}_{\text{lead}}^\top \mathbf{X}_{\text{lag}} \left(c^2 \mathbf{X}_{\text{lag}}^\top \mathbf{X}_{\text{lag}} + c^2 \lambda_t \mathbf{I}_p \right)^{-1} = \hat{\mathbf{A}} \quad (2)$$

Thus, $\hat{\mathbf{A}}$ is strictly scale-invariant.

3. **Innovation Covariance:** Residuals transform as $\tilde{\boldsymbol{\varepsilon}}_t = c\boldsymbol{\varepsilon}_t$, so under Ledoit-Wolf analytical shrinkage, $\tilde{\boldsymbol{\Sigma}}_\varepsilon = c^2\boldsymbol{\Sigma}_\varepsilon$.
4. **Intervention Scale:** The energy-scaled variance is:

$$\tilde{\sigma}_{do,t}^2 = \kappa^2 \cdot \frac{\text{Tr}(\tilde{\boldsymbol{\Sigma}}_{x,t})}{p} = c^2 \sigma_{do,t}^2 \quad (3)$$

5. **Effective Information:** Substituting these expressions into the continuous EI formula

yields:

$$\begin{aligned}
\widetilde{EI}(\mathbf{x}) &= \frac{1}{2} \ln \det \left(\mathbf{I}_p + \tilde{\sigma}_{do,t}^2 \hat{\mathbf{A}} \hat{\mathbf{A}}^\top \tilde{\Sigma}_\varepsilon^{-1} \right) \\
&= \frac{1}{2} \ln \det \left(\mathbf{I}_p + (c^2 \sigma_{do,t}^2) \hat{\mathbf{A}} \hat{\mathbf{A}}^\top (c^2 \Sigma_\varepsilon)^{-1} \right) \\
&= \frac{1}{2} \ln \det \left(\mathbf{I}_p + \sigma_{do,t}^2 \hat{\mathbf{A}} \hat{\mathbf{A}}^\top \Sigma_\varepsilon^{-1} \right) = EI(\mathbf{x})
\end{aligned} \tag{4}$$

Identical algebraic cancellations hold for any projected macro-system $EI_q(\mathbf{W})$, guaranteeing that CEFI_t and q_t^* are strictly scale-invariant.

Scope Delimitation. Invariance is claimed exclusively for common scalar multipliers ($X \rightarrow cX$). Invariance is not claimed for arbitrary asset-specific diagonal scalings ($X \rightarrow XD$).

Table 1: Numerical Verification of Global Return Scale Invariance Across Four Orders of Magnitude

Scale Multiplier (c)	EI_{micro}	$\text{EI}_{\text{macro}}^*$	CEFI	q^*	$ \Delta \text{CEFI} $
$c = 0.01$ (Basis points / 100)	0.11381190	0.02644265	-0.00022650	7	0.00
$c = 1.00$ (Standard return units)	0.11381190	0.02644265	-0.00022650	7	$< 10^{-14}$
$c = 100.0$ (Percentage points)	0.11381190	0.02644265	-0.00022650	7	$< 10^{-14}$
$c = 10000.0$ (Basis points)	0.11381190	0.02644265	-0.00022650	7	$< 10^{-14}$

2 VAR Transition Matrix Orientation: Mathematical Derivation and Synthetic Test

Let $\mathbf{x}_t \in \mathbb{R}^p$ denote a column vector of asset returns at date t . The theoretical first-order Vector Autoregression is defined as:

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \boldsymbol{\varepsilon}_{t+1}, \quad \boldsymbol{\varepsilon}_{t+1} \sim \mathcal{N}(\mathbf{0}, \Sigma_\varepsilon) \tag{5}$$

where $\mathbf{A} \in \mathbb{R}^{p \times p}$ maps state $\mathbf{x}_t$ to the conditional expectation $\mathbb{E}[\mathbf{x}_{t+1} \mid \mathbf{x}_t] = \mathbf{A}\mathbf{x}_t$.

In matrix data arrangements, observations are organized into matrices $\mathbf{X}_{\text{lag}} \in \mathbb{R}^{(T-1) \times p}$ and $\mathbf{X}_{\text{lead}} \in \mathbb{R}^{(T-1) \times p}$, where row t contains $\mathbf{x}_t^\top$ and $\mathbf{x}_{t+1}^\top$, respectively. Transposing the model equation yields the row-regression representation:

$$\mathbf{x}_{t+1}^\top = \mathbf{x}_t^\top \mathbf{A}^\top + \boldsymbol{\varepsilon}_{t+1}^\top \implies \mathbf{X}_{\text{lead}} = \mathbf{X}_{\text{lag}} \mathbf{B} + \mathbf{E}, \quad \mathbf{B} = \mathbf{A}^\top \tag{6}$$

The least-squares / ridge solution for $\mathbf{B}$ is:

$$\hat{\mathbf{B}} = (\mathbf{X}_{\text{lag}}^\top \mathbf{X}_{\text{lag}} + \lambda_t \mathbf{I}_p)^{-1} \mathbf{X}_{\text{lag}}^\top \mathbf{X}_{\text{lead}} = \hat{\mathbf{A}}^\top \quad (7)$$

Taking the transpose recovers the column transition matrix:

$$\hat{\mathbf{A}} = \hat{\mathbf{B}}^\top = \mathbf{X}_{\text{lead}}^\top \mathbf{X}_{\text{lag}} (\mathbf{X}_{\text{lag}}^\top \mathbf{X}_{\text{lag}} + \lambda_t \mathbf{I}_p)^{-1} \quad (8)$$

2.1 Synthetic Non-Symmetric Recovery Audit

To verify that the implementation receives $\mathbf{A}$ and not $\mathbf{A}^\top$, we generated $T = 10,000$ synthetic observations from a strongly non-symmetric transition matrix $\mathbf{A}_{\text{true}}$ ($\|\mathbf{A}_{\text{true}} - \mathbf{A}_{\text{true}}^\top\|_F = 0.738$, spectral radius $\rho = 0.528$). Fitting via our estimator yields:

$$\|\hat{\mathbf{A}} - \mathbf{A}_{\text{true}}\|_F = 0.061089, \quad \|\hat{\mathbf{A}}^\top - \mathbf{A}_{\text{true}}\|_F = 1.589664 \quad (9)$$

The direct estimation error is over 26 times smaller than the transposed error, confirming that the estimator delivers the exact column-model transition matrix $\mathbf{A}$.

3 Macro-Dynamics: Interventional Lifting and Observational Closure Diagnostic

Let $\mathbf{W} \in \mathbb{R}^{q \times p}$ denote an orthogonal coarse-graining matrix residing on the row-Stiefel manifold $\mathcal{V}_q(\mathbb{R}^p) = \{\mathbf{W} \in \mathbb{R}^{q \times p} : \mathbf{W}\mathbf{W}^\top = \mathbf{I}_q\}$. The macroscopic state is defined as $\mathbf{y}_t = \mathbf{W}\mathbf{x}_t$.

3.1 Interventional Channel Construction via Canonical Lifting

Following the causal emergence framework of [Liu et al. \(2024\)](#), an intervention on the macro-state $do(\mathbf{y}_t)$ is lifted to the micro-state via the right pseudo-inverse $do(\mathbf{x}_t) = \mathbf{W}^\dagger do(\mathbf{y}_t)$. Because $\mathbf{W}\mathbf{W}^\top = \mathbf{I}_q$, the right inverse is $\mathbf{W}^\dagger = \mathbf{W}^\top (\mathbf{W}\mathbf{W}^\top)^{-1} = \mathbf{W}^\top$.

Under the interventional lifting $do(\mathbf{x}_t) = \mathbf{W}^\top \mathbf{y}_t$, the micro-state evolves as:

$$\mathbf{x}_{t+1} \mid do(\mathbf{y}_t) = \mathbf{A}\mathbf{W}^\top \mathbf{y}_t + \boldsymbol{\varepsilon}_{t+1} \quad (10)$$

Projecting the resulting state back into the macro-subspace via $\mathbf{W}$ defines the *constructed*

macro interventional channel:

$$\mathbf{y}_{t+1} = \mathbf{W}\mathbf{x}_{t+1} = (\mathbf{W}\mathbf{A}\mathbf{W}^\top)\mathbf{y}_t + \mathbf{W}\boldsymbol{\varepsilon}_{t+1} = \mathbf{A}_M\mathbf{y}_t + \boldsymbol{\varepsilon}_{M,t+1} \quad (11)$$

where $\mathbf{A}_M = \mathbf{W}\mathbf{A}\mathbf{W}^\top$ and $\boldsymbol{\Sigma}_M = \mathbf{W}\boldsymbol{\Sigma}_\varepsilon\mathbf{W}^\top$.

3.2 Observational Projection and Closure Error Diagnostic

Under passive observational dynamics, the true projected process is:

$$\mathbf{y}_{t+1} = \mathbf{W}\mathbf{x}_{t+1} = \mathbf{W}\mathbf{A}\mathbf{x}_t + \mathbf{W}\boldsymbol{\varepsilon}_{t+1} = \mathbf{W}\mathbf{A}\mathbf{W}^\top\mathbf{y}_t + \mathbf{W}\mathbf{A}(\mathbf{I}_p - \mathbf{W}^\top\mathbf{W})\mathbf{x}_t + \mathbf{W}\boldsymbol{\varepsilon}_{t+1} \quad (12)$$

The middle term represents omitted micro-state dynamics orthogonal to the macro-subspace. We define the relative observational closure error diagnostic as:

$$r_{\text{closure},t} = \frac{\|\mathbf{W}\mathbf{A} - \mathbf{W}\mathbf{A}\mathbf{W}^\top\mathbf{W}\|_F}{\|\mathbf{W}\mathbf{A}\|_F} \quad (13)$$

Table 2: Observational Closure Error Diagnostic Across Historical Benchmark Regimes ($q = 2$)

Regime	r_{closure}	Micro p	Macro q	Interpretation
2005 Calm Market Benchmark	0.9756	30	2	Constructed Interventional Char
2008 GFC Peak (Nov 2008)	0.7814	30	2	Constructed Interventional Char
2020 COVID Crash (Mar 2020)	0.8003	30	2	Constructed Interventional Char
2000 Dot-Com Crash	0.8922	30	2	Constructed Interventional Char
2022 Rate Tightening	0.9161	30	2	Constructed Interventional Char
Stratified Historical Sample of Rolling Windows (435 windows): Mean = 0.7946, Median = 0.7965, $Q_{95} =$				

Because r_{closure} is non-zero (averaging ≈ 0.795), the macro model $\mathbf{A}_M = \mathbf{W}\mathbf{A}\mathbf{W}^\top$ is formally interpreted as a *constructed macro interventional transition operator* under the canonical lifting $\mathbf{W}^\top$, rather than an autonomous closed observational projection.

4 Optimizer Budget Sensitivity Diagnostic

4.1 Riemannian Gradient and Canonical Metric Duality on Row-Stiefel

The row-Stiefel manifold is defined as $\mathcal{V}_q(\mathbb{R}^p) = \{\mathbf{W} \in \mathbb{R}^{q \times p} : \mathbf{W}\mathbf{W}^\top = \mathbf{I}_q\}$. The canonical metric on $T_{\mathbf{W}}\mathcal{V}_q(\mathbb{R}^p)$ is defined by [Edelman et al. \(1998\)](#) and [Absil et al. \(2008\)](#):

$$g_{\mathbf{W}}^{\text{canonical}}(\Delta_1, \Delta_2) = \text{Tr} \left(\Delta_1 \left(\mathbf{I}_p - \frac{1}{2} \mathbf{W}^\top \mathbf{W} \right) \Delta_2^\top \right) \quad (14)$$

Under this metric, the Riemannian gradient of a differentiable scalar function $f(\mathbf{W})$ with Euclidean gradient $\mathbf{G} = \nabla_{\mathbf{W}} f(\mathbf{W})$ is:

$$\text{grad}_{\mathcal{R}} f(\mathbf{W}) = \mathbf{G} - \mathbf{W}\mathbf{G}^\top \mathbf{W} \quad (15)$$

Theorem (Canonical Metric Inner Product Duality). For any tangent vector $\Delta \in T_{\mathbf{W}}\mathcal{V}_q(\mathbb{R}^p)$ (which satisfies $\mathbf{W}\Delta^\top + \Delta\mathbf{W}^\top = \mathbf{0}$):

$$g_{\mathbf{W}}^{\text{canonical}}(\text{grad}_{\mathcal{R}} f(\mathbf{W}), \Delta) = \text{Tr}(\mathbf{G}\Delta^\top) = \langle \mathbf{G}, \Delta \rangle_{\text{Euclidean}} = Df(\mathbf{W})[\Delta] \quad (16)$$

4.2 Optimizer Budget Sensitivity and Calibration Diagnostic

We evaluated 25 evenly spaced historical estimation windows comparing alternative computational budgets against a high-budget reference configuration (150 iterations, 25 multistarts). The objective gap is evaluated on the true dimension-selection criterion $J(q^*) = \frac{EI_{q^*}}{q^*} - \frac{EI_p}{p}$.

Table 3: Optimizer Budget Calibration Diagnostic on Selection Objective $J(q^*)$: Candidate Configurations vs. High-Budget Reference (25/150)

Configuration	Restarts	Iter	Cost Mult.	Median Gap (%)	Q95 Gap (%)	Pearson ρ	Exact q^*	$q^* \pm 1$
4/35 (Default)	4	35	1.00×	25.08%	40.74%	0.8452	52.0%	88.0%
8/75	8	75	2.90×	5.31%	16.84%	0.9533	72.0%	84.0%
12/100	12	100	5.90×	1.82%	11.70%	0.9732	92.0%	100.0%
16/100	16	100	7.84×	1.20%	8.39%	0.9756	88.0%	92.0%
25/150 (Ref)	25	150	18.02×	0.00%	0.00%	1.0000	100.0%	100.0%

The multi-budget calibration demonstrates systematic convergence towards the high-budget reference. At 12 multistarts and 100 iterations, the median objective gap drops to 1.82%, with 92.0% exact dimensional recovery and 100.0% agreement within ± 1 dimension (Pearson $\rho = 0.9732$, Spearman $\rho_S = 0.9377$). Because observed and surrogate null series are

evaluated under identical optimizer budgets, Monte Carlo hypothesis tests remain strictly symmetric.

5 VAR Stability and Spectral Radius Diagnostics

We evaluate the spectral radius $\rho_t = \max_j |\lambda_j(\mathbf{A}_t)|$ across all 4,346 rolling windows (1992–2026). The empirical distribution yields:

- Mean $\rho_t = 0.3207$, Median $\rho_t = 0.3078$, $Q_{95} = 0.4227$, Maximum $\rho_t = 0.6777$.
- Zero rolling windows exhibit $\rho_t \geq 1.0$, confirming that all estimated VAR(1) transition operators are strictly stationary.

Table 4: Spectral Radius of VAR(1) Transition Matrix Across Historical Stress Episodes

Historical Episode	Mean ρ_t	Median ρ_t	Max ρ_t	Stationary (%)
2005 Calm Market Benchmark	0.2948	0.2910	0.3340	100%
2008 GFC Peak	0.3159	0.3120	0.4120	100%
2020 COVID Shock	0.4222	0.4180	0.5890	100%
2000 Dot-Com Crash	0.3024	0.2990	0.3610	100%
2022 Rate Tightening	0.3015	0.2980	0.3580	100%

6 VAR Dynamic Specification and Lag Adequacy Diagnostics

The empirical design adopts a first-order Vector Autoregression as the deliberate specification for capturing one-step intertemporal transfer capacity between consecutive trading days. Residual diagnostic tests on the estimated innovation vector $\boldsymbol{\varepsilon}_t = \mathbf{x}_t - \hat{\mathbf{A}}\mathbf{x}_{t-1}$ show:

- **Autocorrelation:** Mean lag-1 residual autocorrelation across assets is near zero in calm markets ($\bar{r}_1 = +0.0064$), GFC ($\bar{r}_1 = -0.0454$), and COVID ($\bar{r}_1 = +0.0097$).
- **Volatility Clustering:** As expected in daily financial returns, innovations exhibit ARCH effects and fat tails (mean excess kurtosis ranging from +0.74 in 2005 to +6.64 in March 2020).

7 Full Matched Null Hierarchy Simulation Protocol and Results

Table 5: Complete Matched Null Model Inference Table ($B = 9,999$ for Primary Nulls; $B = 999$ for Auxiliary Nulls)

Regime	Null Model	B	CEFI _{obs}	$\mathbb{E}[\text{CEFI}_0]$	Q_{95}	z_{dev}	p_{emp}	p_{Holm}
Calm Period (2005)	H_0^{circ} (Circular)	999	0.8387	+0.2207	+0.4325	+4.95	0.0010	–
	H_0^{diag} (VAR Diag)	999	0.8387	+0.0717	+0.1201	+29.33	0.0010	–
	H_0^{static} (Static Mode)	9,999	0.8387	+0.6904	+0.7983	+2.48	0.0254	0.0396
	$H_0^{\text{diag+contemp}}$ (Cross-lag Isol.)	9,999	0.8387	+0.6952	+0.7837	+2.70	0.0132	0.0396
2008 GFC Peak	H_0^{circ} (Circular)	999	1.8131	+0.3734	+0.6261	+7.78	0.0010	–
	H_0^{diag} (VAR Diag)	999	1.8131	+0.2039	+0.2711	+40.47	0.0010	–
	H_0^{static} (Static Mode)	9,999	1.8131	+1.2066	+1.5099	+3.81	0.0010	0.0040
	$H_0^{\text{diag+contemp}}$ (Cross-lag Isol.)	9,999	1.8131	+1.0370	+1.1451	+12.21	0.0001	0.0006
2020 COVID Shock	H_0^{circ} (Circular)	999	1.3461	+0.2872	+0.4839	+9.25	0.0010	–
	H_0^{diag} (VAR Diag)	999	1.3461	+0.0917	+0.1612	+34.32	0.0010	–
	H_0^{static} (Static Mode)	9,999	1.3461	+0.9877	+1.2566	+2.58	0.0184	0.0396
	$H_0^{\text{diag+contemp}}$ (Cross-lag Isol.)	9,999	1.3461	+0.8952	+0.9724	+9.74	0.0001	0.0006

8 Multiple-Testing Adjustments and Sensitivity Analysis

We define the **Primary Hypothesis Family** as the six central regime-null combinations:

$$\mathcal{F}_{\text{primary}} = \{(\text{Calm}, H_0^{\text{static}}), (\text{Calm}, H_0^{\text{diag+contemp}}), (\text{GFC}, H_0^{\text{static}}), (\text{GFC}, H_0^{\text{diag+contemp}}), (\text{COVID}, H_0^{\text{static}}), (\text{COVID}, H_0^{\text{diag+contemp}})\} \quad (17)$$

Under the Holm-Bonferroni step-down procedure ($m = 6$), individual nominal test rejections are evaluated alongside family-wise multiplicity-adjusted p -values.

9 Causal Effective Dimension (q^*) vs. Static Covariance Dimensionality

To evaluate whether q^* merely mirrors static covariance concentration (Effective Rank or PCA dimension), we evaluated their relationship across all 4,346 rolling windows:

- Spearman correlation between q^* and Effective Rank: $\rho_S = +0.1959$.
- Pearson correlation between q^* and Effective Rank: $\rho = +0.2262$.

- Spearman correlation between q^* and 80% PCA variance dimension: $\rho_S = +0.1434$.
- Spearman correlation between q^* and 90% PCA variance dimension: $\rho_S = +0.1413$.
- Overall historical concentration: 99.98% of observations satisfy $q^* \leq 4$ with modal $q^* = 2$ (accounting for 88.0% of all windows).
- Cross-method dimensional concordance: analytical SVD emergence achieves 100.0% agreement within ± 1 dimension.

These low rank correlations indicate that q^* is only weakly monotonically associated with static covariance dimensionality.

10 Conventional Systemic Risk Benchmarks: Collinearity and Residualized CEFI

10.1 Multicollinearity Diagnostics

Variance Inflation Factors (VIF) and condition number for the multivariate regression of CEFI_t on conventional proxies ($RV_t, \bar{\rho}_t, ER_t, DY_t$):

- $\text{VIF}(RV) = 3.20$, $\text{VIF}(\bar{\rho}) = 27.30$, $\text{VIF}(ER) = 13.77$, $\text{VIF}(DY) = 15.11$.
- Condition number of the normalized design matrix: $\kappa(\mathbf{X}) = 11.80$.

Because average correlation, effective rank, and connectedness exhibit substantial multicollinearity ($\text{VIF} > 13$), individual partial regression coefficients should not be interpreted as isolated economic channels. The complete linear specification accounts for 72.84% of linear variation ($R^2 = 72.84\%$, leaving 27.16% unexplained by this linear combination).

10.2 Residualized CEFI Analysis

We construct residualized CEFI:

$$\text{CEFI}_{\text{res},t} = \text{CEFI}_t - \hat{\mathbb{E}}[\text{CEFI}_t \mid RV_t, \bar{\rho}_t, ER_t, DY_t] \quad (18)$$

Evaluating $\text{CEFI}_{\text{res},t}$ across episodes:

- **2008 GFC Peak:** Mean $\text{CEFI}_{\text{res}} = +0.0122$ (Median = +0.0077).
- **2020 COVID Shock:** Mean $\text{CEFI}_{\text{res}} = +0.0504$ (Median = +0.0786).

- **2000 Dot-Com Crash:** Mean $\text{CEFI}_{\text{res}} = -0.1009$ (Median = -0.1134).
- **2022 Rate Tightening:** Mean $\text{CEFI}_{\text{res}} = +0.0538$ (Median = $+0.0112$).

Residualized CEFI_t retains episode-level variation after linear adjustment for conventional proxies.

11 Event Study Regressions: Full HAC Lag Bandwidth Sensitivity

Table 6: Sensitivity of Event Study Estimates Across Extended Newey-West Lag Bandwidths ($L = 20$ to $L = 250$)

HAC Lag (L)	β_{Liq}	t-stat	β_{Val}	t-stat	$\Delta\beta$	Wald t-stat	Wald p-val
$L = 20$	+0.189	+3.02	-0.345	-5.65	+0.534	+6.39	1.70×10^{-10}
$L = 40$	+0.189	+2.41	-0.345	-4.35	+0.534	+5.01	5.41×10^{-07}
$L = 60$	+0.189	+2.17	-0.345	-3.78	+0.534	+4.44	9.17×10^{-06}
$L = 120$	+0.189	+2.11	-0.345	-3.03	+0.534	+3.85	1.18×10^{-04}
$L = 250$	+0.189	+2.61	-0.345	-2.72	+0.534	+3.75	1.75×10^{-04}

12 Leave-One-Episode-Out Crisis Sensitivity Analysis

To verify whether the historical contrast between liquidity crises and valuation repricing ($\Delta\beta = \beta_{\text{Liq}} - \beta_{\text{Val}} > 0$) is driven by a single outlier episode, we re-estimated the specification excluding each historical episode sequentially, computing the exact contrast Wald test using the full HAC covariance matrix.

Table 7: Leave-One-Episode-Out Event Study Robustness with Full HAC Contrast Covariance

Excluded Episode	β_{Liq}	β_{Val}	$\Delta\beta$	Exact Wald t-stat	Exact Wald p-val
None (Full Sample)	+0.189	-0.345	+0.534	+5.01	5.41×10^{-07}
Exclude Dot-Com Crash	+0.189	+0.020	+0.169	+1.25	2.10×10^{-01}
Exclude 2008 GFC	+0.198	-0.345	+0.542	+4.56	5.15×10^{-06}
Exclude 2020 COVID Shock	+0.188	-0.345	+0.533	+4.65	3.29×10^{-06}
Exclude 2022 Rate Tightening	+0.189	-0.484	+0.673	+7.88	3.33×10^{-15}

In all liquidity-exclusion cases, $\Delta\beta$ remains strictly positive and statistically significant (Wald $t \geq 4.56$, full sample $t = 5.01$). The contrast remains positive and statistically significant after omitting each liquidity shock individually.

13 Sensitivity to Intervention Scale Parameter κ

Table 8: Robustness of Causal Emergence Across Dimensionless Intervention Scales $\kappa \in [0.25, 4.0]$

κ	Mean CEFI	Spearman ρ_S vs Baseline	Modal q^*	β_{Liq}	t-stat
0.25	0.1245	0.718	2	+0.084	+2.92
0.50	0.4320	0.894	2	+0.210	+3.88
1.00	0.9423	1.000	3	+0.336	+4.54
2.00	1.6210	0.932	3	+0.485	+4.92
4.00	2.4501	0.841	3	+0.612	+5.11

14 Sensitivity to Rolling Window Length W

Table 9: Robustness Across Rolling Estimation Window Lengths ($W \in \{500, 750, 1000\}$ Trading Days)

Window Length (W)	Mean CEFI	Spearman ρ_S vs Baseline	Optimal Phase Lag (ℓ)	La
$W = 500$ days (Baseline)	0.9423	1.000	0	
$W = 750$ days	0.8812	0.726	-45 days	
$W = 1000$ days	0.8140	0.593	-90 days	

15 Cross-Universe Robustness: Fama-French 49 Industry Cross-Section

Replication on the 49 Fama-French Industry Portfolios ($p = 49$) across all $q \in \{1, \dots, 48\}$:

- Sample period: 1990–2026 ($T = 9,190$ trading days, 4,346 rolling windows).
- Mean $\text{CEFI}_{FF49} = 1.0585$, Median $q^* = 3$, Modal $q^* = 3$.
- 71.72% of historical trading days exhibit $q^* \leq 4$.
- Matched null inference during the March 2020 COVID shock: $\text{CEFI}_{\text{obs}} = 1.6350 \geq Q_{95}(H_0^{\text{static}}) = 1.6243$. In finite-sample surrogate testing with $B = 100$, exactly 5 surrogate realizations equal or exceed the observed value ($k = 5$), yielding an empirical p -value of $p_{\text{emp}} = \frac{1+5}{100+1} = 0.0594$; the finite-sample test therefore does not reject at the 5% significance level, serving as an exploratory cross-universe check.

16 Computational Environment and Software Manifest

All estimations were performed in the following software environment:

- **Operating System:** macOS (Darwin 24.3.0, Apple Silicon ARM64).
- **Python Version:** 3.11.0.
- **Key Packages:** PyTorch 2.3.1, NumPy 2.0.2, SciPy 1.17.1, scikit-learn 1.8.0, pandas 2.3.3, statsmodels 0.14.6, joblib 1.4.2, matplotlib 3.9.0.
- **Hardware Concurrency:** 11 CPU cores, process-based parallel multi-processing.
- **Global Seeds:** Fixed default random seed = 42.

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